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MEMS element and vibration-driven energy harvesting device

Abstract

A MEMS element according to the present invention is provided with a base, an insulation layer fixed to one surface of the base, a first upper layer at least portions of which are fixed to the insulation layer, and a second upper layer provided surrounding the first upper layer and disposed being separated from the first upper layer by slits, wherein the first upper layer includes, at predetermined portions, protruding portions protruding toward the second upper layer, and the protruding portions are fixed to the insulation layer.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a MEMS element and a vibration-driven energy harvesting

device.

BACKGROUND ART

(2) MEMS elements which are produced on a silicon on insulator (SOI) substrate by forming fixed portions including fixed comb teeth and a moveable portion including movable comb teeth on an insulation layer made of a silicon oxide layer or the like formed on a silicon substrate are known. At least portions of each of the fixed portions and the moveable portion are fixed to the insulation layer. The fixed portions are each formed with a lead portion connected to an electrode terminal. In some configuration, an outer peripheral fixed pattern is formed around the fixed portions including the lead portions.

(3) The fixed portions, the moveable portion, and the outer peripheral fixed pattern are separated from one another by narrow slits formed by etching and are electrically insulated from one another. When the slits are formed by etching, fixed spots fixed to the insulation layer are undercut, and thus, for example, portions having a narrow width at its fixed spots, such as the lead portions, become insufficient in a strength of bonding, so as to be suspended over the insulation layer or broken at a fixed spot by a slight impact.

(4) In light of such circumstances, coping with a decrease in strength of the insulation layer by undercutting is demanded.

(5) There is known a method that does not deal with lead portions but relates to a method for forming slits between fixed portions and a moveable portion. In this method, slits are formed, protective films are then formed on lateral surfaces of the fixed portions, and an insulation layer is removed. In this manner, undercutting of the insulation layer under the fixed portions is prevented, and a strength of bonding of the insulation layer under the fixed portions is kept. This method will be described below.

(6) First, a photoresist is applied to the entire surfaces of the fixed portions and the moveable portion including slits, this photoresist is patterned, and the photoresist on other than surrounding lateral surfaces of the fixed portions is removed by RIE (Reactive Ion Etching) or the like. Next, ashing is performed with oxygen plasma or the like to leave the photoresist on only lateral surfaces of the fixed portions facing the moveable portion. When wet etching using etchant is performed thereafter, the left photoresist serves as protective films against the etching, the etching is prevented on portions where the protective films are formed, so that undercutting of lower portions of the fixed portions and the insulation layer can be prevented (e.g., see Patent Literature 1).

CITATION LIST

Patent Literature

(7) Patent Literature 1: Japanese Patent Laid-Open No. 2005-323039 Patent Literature 2: Japanese Patent Laid-Open No. 2018-88780

SUMMARY OF INVENTION

Technical Problem

(8) The method described in Patent Literature 1 has poor productivity, increasing costs because its processing process is long and complicated.

Solution to Problem

(9) A MEMS element according to a first aspect of the present invention includes a base, an insulation layer fixed to one surface of the base, a first upper layer at least portions of which are fixed to the insulation layer, and a second upper layer provided surrounding the first upper layer and disposed being separated from the first upper layer by slits, the first upper layer includes protruding portions protruding toward the second upper layer, and the protruding portions are fixed to the insulation layer.

(10) According to a second aspect of the present invention, in the MEMS element according to the first aspect, the first upper layer preferably includes lead portions, and the protruding portions are preferably provided on the lead portions.

(11) According to a third aspect of the present invention, in the MEMS element according to the

first or second aspect, the insulation layer is preferably removed in regions corresponding to the slits, and the base is preferably exposed in a thickness direction of the insulation layer through at least some of the regions corresponding to the slits.

(12) According to a fourth aspect of the present invention, in the MEMS element according to any one of the first to third aspects, the second upper layer is preferably fixed to the insulation layer.

(13) According to a fifth aspect of the present invention, in the MEMS element according to any one of the first to fourth aspects, the base, the first upper layer, and the second upper layer are preferably made of silicon.

(14) According to a sixth aspect of the present invention, in the MEMS element according to any one of the first to fifth aspects, the insulation layer is preferably made of an inorganic insulation material.

(15) According to a seventh aspect of the present invention, in the MEMS element according to any one of the first to sixth aspects, the first upper layer preferably includes a plurality of fixed comb teeth, and the insulation layer is preferably removed in regions corresponding to between the fixed comb teeth formed adjacently.

(16) A vibration-driven energy harvesting device according to an eighth aspect of the present invention includes the MEMS element according to the seventh aspect, a moveable portion including a plurality of movable comb teeth meshing with the fixed comb teeth of the first upper layer, elastically-supporting portions elastically supporting the moveable portion, and an output portion outputting electric power that is generated by movement of electric charge occurring by relative movement between the fixed comb teeth and the movable comb teeth.

Advantageous Effects of Invention

(17) According to the present invention, it is possible to prevent a decrease in a strength of fixing the first upper layer and to improve productivity.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a plan view of a vibration-driven energy harvesting device, which is penetrated an upper cover, with a MEMS element enclosed in a vacuum package.

(2) FIG. 2 is a cross-sectional view of the vibration-driven energy harvesting device illustrated in FIG. 1 taken along the line II-II.

(3) FIG. 3(A) is a plan view of the MEMS element illustrated in FIG. 1, and FIG. 3(B) is a plan view illustrating a state where fixed electrode portions and movable electrode portion are removed from the MEMS element illustrated in FIG. 3(A).

(4) FIG. 4(A) to FIG. 4(D) are diagrams illustrating an example of a production method for the MEMS element illustrated in FIG. 3.

(5) FIG. 5(A) to FIG. 5(D) are diagrams illustrating the example of the production method for the MEMS element subsequent to FIG. 4.

(6) FIG. 6 is an enlarged view of a region VI in FIG. 1.

(7) FIG. 7(A) is a cross-sectional view taken along the line VIIA-VIIA in FIG. 6, and FIG. 7(B) is a cross-sectional view of a structure of a region that corresponds to FIG. 7(A) in a comparative example.

DESCRIPTION OF EMBODIMENTS

(8) An embodiment for carrying out the present invention will be described below with reference to the drawings.

(9) FIG. 1 is a plan view of a vibration-driven energy harvesting device 1, which is penetrated an upper cover 3, with a MEMS element 10 enclosed in a vacuum package, and FIG. 2 is a cross-sectional view taken along the line II-II in FIG. 1.

(10) The case **2** and the upper cover **3** forms the vacuum package, and the MEMS element **10** is housed in this vacuum package. In the plan view of FIG. **1**, the upper cover **3** provided on an upper side (a positive z-axis direction side) is not illustrated for illustrating a planar structure of the MEMS element **10** clearly.

(11) Note that, in the present embodiment, an x-axis direction, a y-axis direction, and the z-axis direction are supposed to be the respective directions illustrated in each drawing.

(12) The MEMS element **10** includes four fixed electrode portions (first upper layer) **11**, a fixed-electrode outer periphery portion (second upper layer) **35** surrounding the fixed electrode portions **11**, a movable electrode portion (moveable portion) **12**, and elastically-supporting portions **13** elastically supporting the movable electrode portion **12**. As illustrated in FIG. **2**, a base **7** of the MEMS element **10** is fixed to the case **2** by die bonding. The case **2** is formed of, for example, an electrical insulation material (e.g., ceramic). To an upper end of the case **2**, the upper cover **3** for vacuum sealing the inside the case **2** is seam welded.

(13) As illustrated in FIG. **2**, the MEMS element **10** includes the base **7** made of Si, a device layer **9** made of a Si active layer, and an insulation layer **8** made of an inorganic insulation material, such as SiO₂, that bonds the base **7** and the device layer **9** together. That is, the MEMS element **10** is configured as a three-layered structure in which the base **7**, the insulation layer **8**, and the device layer **9** made of the Si active layer are stacked in the z-axis direction. The MEMS element **10** having such a configuration is usually formed from SOI (Silicon On Insulator) substrate by a general MEMS process technique.

(14) The device layer **9** includes the four fixed electrode portions **11**, the fixed-electrode outer periphery portion **35**, the movable electrode portion **12**, and the elastically-supporting portions **13**. The fixed electrode portions **11** each include a plurality of fixed comb teeth **110**, a fixed-comb-teeth connecting portion **111** connecting the plurality of fixed comb teeth **110**, and a lead portion **112**. The fixed comb teeth **110** are made to extend in the x-axis direction and are arranged in the y-axis direction at predetermined intervals. The fixed-comb-teeth connecting portions **111** are made to extend in the y-axis direction and each connect the plurality of fixed comb teeth **110** arranged in the y-axis direction. The lead portions **112** are made to extend in a direction perpendicular to the fixed-comb-teeth connecting portions **111**, that is, in the x-axis direction. The end of the lead portions **112** includes an end portion that is formed with a terminal portion having a rectangular shape. On an upper surface of this terminal portion, a conductive metal such as aluminum is provided and formed as an electrode pad **113**.

(15) At a predetermined spot on one lateral surface of the lead portion **112**, which extends in the x-axis direction (see a region VI), a protruding portion **15** protruding toward the fixed-electrode outer periphery portion **35** (see FIG. **6**) is provided. The protruding portion **15** provided on the lead portion **112** will be described later.

(16) Between the fixed-electrode outer periphery portion **35**, and the lead portion **112** and the fixed-comb-teeth connecting portion **111** of each fixed electrode portion **11**, a slit **16** is provided, by which the fixed-electrode outer periphery portion **35** is physically separated from the lead portion **112** and the fixed-comb-teeth connecting portion **111** of each fixed electrode portion **11**. In this configuration, the fixed-electrode outer periphery portion **35** and each fixed electrode portion **11** are electrically insulated from each other. The lead portion **112** and the fixed-comb-teeth connecting portion **111** of each fixed electrode portion **11** are supported by the base **7** with the insulation layer **8** interposed therebetween. The fixed comb teeth **110** of each fixed electrode portion **11** are extended over a region corresponding to a rectangular shaped opening **7a** (see FIG. **2** and FIGS. **3(A)** and **3(B)**) in the base **7**.

(17) The movable electrode portion **12** includes a plurality of movable comb teeth **120**, a center band portion **121**, and a movable-comb-teeth connecting portions **122** connecting the plurality of movable comb teeth **120**. The movable-comb-teeth connecting portions **122** are made to extend from a center of the center band portion **121** in the x-axis direction to positive and negative y-axis

directions. The movable comb teeth **120** are made to extend in positive and negative x-axis directions from the movable-comb-teeth connecting portions **122** made to extend in the positive and negative y-axis directions and are arranged in the y-axis direction at predetermined intervals.

(18) As illustrated in FIG. 2, weights **105a** and **105b** are fixed to upper and lower surfaces of the center band portion **121** of the movable electrode portion **12**, which are surfaces on positive and negative z-axis direction sides of the center band portion **121**, respectively, by bonding or the like. Positions of gravity centers of the weights **105a** and **105b** are coaxial with a z-axis that passes through centers of the center band portion **121** in the x-axis direction and the y-axis direction.

(19) Two of the fixed electrode portions **11** disposed on the positive y-axis direction side of the center band portion **121** are disposed to have a line symmetry with respect to a center line of the center band portion **121** in the x-axis direction. The other two of the fixed electrode portions **11** disposed on the negative y-axis direction side of the center band portion **121** are disposed to have a line symmetry with respect to the center line of the center band portion **121** in the x-axis direction.

(20) The plurality of fixed comb teeth **110** extending in the x-axis direction from the fixed-comb-teeth connecting portions **111** and the movable comb teeth **120** made to extend in the x-axis direction from the movable-comb-teeth connecting portions **122** are disposed such that the fixed comb teeth **110** and the movable comb teeth **120** mesh with each other in the y-axis direction with gaps interposed therebetween.

(21) The movable electrode portion **12** is mechanically and electrically connected to a vibration regulating portions **150**, which is fixed to the base **7** via the insulation layer **8**, via the elastically-supporting portions **13**. The vibration regulating portions **150** are provided one by one on the positive and negative x-axis direction sides of the center band portion **121**, that is, in a pair. The pair of vibration regulating portions **150** are formed into the same shape and are disposed to have line symmetries with respect to central axes of the center band portion **121** in the x-axis direction and the y-axis direction.

(22) The movable electrode portion **12** supported by the elastically-supporting portions **13** vibrates in the x-axis direction by vibration from the outside, and one side surface **121a** of the center band portion **121** of the movable electrode portion **12** collides with a projection **151** of the vibration regulating portion **150**. At this time, if a position in the y-axis direction of the projection **151** of the vibration regulating portion **150** with which the moveable portion collides deviates in the y-axis direction from a central axis passing through a gravity center of the center band portion **121** including the weights **105a** and **105b**, a moment occurs in the center band portion **121** of the movable electrode portion **12**. When a moment occurs in the center band portion **121** of the movable electrode portion **12**, the elastically-supporting portion **13** deforms, not allowing the center band portion **121** to vibrate normally. It is therefore necessary for a center line in the y-axis direction of contact portions of the vibration regulating portions **150** with which the center band portion **121** of the movable electrode portion **12** collides to be coaxial with the center line of the center band portion **121** of the movable electrode portion **12** extending in the x-axis direction.

(23) To the vibration regulating portions **150**, electrode pads **114** are connected. The vibration regulating portions **150** are each formed integrally with a terminal portion having a rectangular shape, and on an upper surface of this terminal portion, a conductive metal such as aluminum is provided and formed as the electrode pad **114**.

(24) The electrode pads **113** and **114** are connected to electrodes **21a** and **21b** provided on the case **2** by wires **22**, respectively.

(25) The fixed electrode portions **11** and the movable electrode portion **12** are each formed with electrets. In a case where only one of either the fixed electrode portions **11** or the movable electrode portion **12** is formed with electrets, an electric charge of the reversed polarity is produced in the other, and therefore, only one of either the fixed electrode portions **11** or the movable electrode portion **12** may be formed with electrets.

(26) In the present embodiment, the movable electrode portion **12** is adapted to vibrate in the x-axis

direction, and when the movable electrode portion **12** vibrates in the x-axis direction, a degree of insertion of the movable comb teeth **120** of the movable electrode portion **12** relative to the fixed comb teeth **110** of the fixed electrode portions **11** changes, causing movement of an electric charge, by which electric power generation is performed.

(27) FIG. 3(A) is a diagram illustrating the MEMS element **10** before the weights **105a** and **105b** are fixed thereto.

(28) As described above, the MEMS element **10** is formed from SOI (Silicon On Insulator) substrate by a general MEMS process technique. An SOI substrate is configured to have a three-layered structure in which the base **7**, the insulation layer **8**, and the device layer **9** made of a Si active layer are stacked in the z-axis direction. As illustrated in FIG. 2, the device layer **9** is supported by the base **7** via the insulation layer **8**. The fixed electrode portions **11**, the fixed-electrode outer periphery portion **35**, the movable electrode portion **12**, the elastically-supporting portions **13**, and the vibration regulating portions **150** are each formed of a Si active layer.

(29) In FIG. 3(A), the fixed electrode portions **11**, the movable electrode portion **12**, the elastically-supporting portions **13**, and the vibration regulating portions **150** on the base **7** are illustrated by hatching them. The movable electrode portion **12** is elastically supported by four elastically-supporting portions **13**. The elastically-supporting portions **13** each include three beams **13a** to **13c** elastically deformable. The movable electrode portion **12** is disposed in a region that corresponds to the opening **7a** (see FIG. 3(B)) provided in the base **7**. Via the beams **13a** to **13c** of the elastically-supporting portions **13**, the movable electrode portion **12** is connected to the vibration regulating portions **150**. The vibration regulating portions **150** are fixed to the base **7** via the insulation layer **8**. As a result, the movable electrode portion **12** is supported by the base **7** via the four elastically-supporting portions **13** and the vibration regulating portions **150**.

(30) The vibration regulating portions **150** also function as restricting portions that restrict a vibration range of movable electrode portion **12** in the x-axis direction. The vibration of the movable electrode portion **12** in the x-axis direction is restricted by collision of the movable electrode portion **12** with the projection **151** of each vibration regulating portion **150**.

(31) FIG. 3(B) is a plan view illustrating a state where the fixed electrode portions and the movable electrode portion are removed from the MEMS element illustrated in FIG. 3(A).

(32) Hatched zones **11C** in FIG. 3(B) illustrate a pattern of bonding portions at which the fixed-comb-teeth connecting portions **111** and the lead portions **112** of the fixed electrode portions **11** are bonded to the insulation layer **8**. Hatched zones **11A** in FIG. 3(B) illustrate a pattern of bonding portions at which end portions of the beams **13a** of the elastically-supporting portions **13** are bonded to the insulation layer **8**. Hatched zones **11B** in FIG. 3(B) illustrate a pattern of bonding portions at which the vibration regulating portions **150** are bonded to the insulation layer **8**.

(33) Next, a producing method for the MEMS element **10** will be described.

(34) FIG. 4(A) to FIG. 4(D) are diagrams illustrating an example of the production method for the MEMS element illustrated in FIG. 3, and FIG. 5(A) to FIG. 5(D) are diagrams illustrating the example of the production method for the MEMS element subsequent to FIG. 4.

(35) Note that FIG. 4 and FIG. 5 schematically illustrate cross sections taken along the dash-dot line C-C in FIG. 3(A).

(36) FIG. 4(A) is a diagram illustrating a cross section of the SOI substrate that forms the MEMS element **10**. The SOI substrate includes a base layer **301** made of Si, an insulation layer **302** made of SiO₂, and a device layer **303** made of a Si active layer. In a first step illustrated in FIG. 4(B), a nitride film (SiN film) **304** is deposited on a surface of the device layer **303**. In a second step illustrated in FIG. 4(C), the nitride film **304** is patterned into a nitride film pattern **304a** for protecting spots at which the electrode pads **113** and **114** are to be formed.

(37) In a third step illustrated in FIG. 4(D), a mask pattern for forming the fixed electrode portions **11**, the movable electrode portion **12**, the elastically-supporting portions **13**, and the vibration regulating portions **150** is formed and disposed on the device layer **303**, and the device layer **303** is

etched. The etching process is performed by DRIE (Deep Reactive Ion Etching) or the like until the insulation layer **302** is reached. In FIG. **4(D)**, an area indicated by reference character **B1** is an area that corresponds to the fixed electrode portion **11**, an area indicated by reference character **B2** is an area that corresponds to the movable electrode portion **12**, and an area indicated by reference character **B3** is an area that corresponds to the vibration regulating portion **150**.

(38) In the cross section taken along the dash-dot line C-C in FIG. **3(A)**, the fixed electrode portion **11** indicated by reference character **B1** includes the electrode pad **113** and the fixed-comb-teeth connecting portion **111**, the movable electrode portion **12** indicated by reference character **B2** includes the movable-comb-teeth connecting portion **122** and the center band portion **121**, and the area indicated by reference character **B3** includes the vibration regulating portion **150** and the electrode pad **114**. Note that illustration of the electrode pad **114** is omitted.

(39) In addition, the fixed-electrode outer periphery portion **35** is formed on the outside of the electrode pad **113** in the negative x-axis direction and between the electrode pad **113** and the fixed-comb-teeth connecting portion **111**. The slit **16** is formed between the outside of the electrode pad **113** and the electrode pad **113**, between the electrode pad **113** and the fixed-electrode outer periphery portion **35**, and between the fixed-electrode outer periphery portion **35** and the fixed-comb-teeth connecting portion **111**.

(40) In a fourth step illustrated in FIG. **5(A)**, a mask pattern for forming the opening **7a** in the base **7** is formed on a lower surface of the base layer **301**, and the base layer **301** is processed by the DRIE. The opening **7a** is thereby formed in the base layer **301**, and the base layer **301** is formed into the base **7** having the opening **7a**. In a fifth step illustrated in FIG. **5(B)**, the insulation layer **302** made of SiO₂ that is exposed through the opening **7a** of the base layer **301** is removed by strong hydrofluoric acid.

(41) Note that, in FIG. **5(B)**, only the fixed-comb-teeth connecting portion **111** that is a portion of the fixed electrode portion **11** and only the movable-comb-teeth connecting portion **122** and the center band portion **121** that are portions of the movable electrode portion **12** are illustrated. Although the movable-comb-teeth connecting portion **122** and the center band portion **121** are illustrated in FIG. **5(B)** in a state of floating over the base layer **301**, the movable electrode **12** is supported via the vibration regulating portions **150** fixed to portions of the insulation layer **8** illustrated as the hatched zones **11B** and via the elastically-supporting portions **13** connected to the vibration regulating portions **150**, as illustrated in FIG. **3(B)**.

(42) In a sixth step illustrated in FIG. **5(C)**, silicon dioxide films **305** are formed on surfaces of the base layer **301** and the device layer **303** by a thermal oxidation process. At this time, the silicon dioxide films **305** are also formed on lateral surfaces of the fixed electrode portion **11** in the slit **16** and lateral surfaces of the fixed-electrode outer periphery portion **35** in the slit **16**. In a sixth step illustrated in FIG. **5(D)**, the nitride film pattern **304a** is removed, and an aluminum electrode **113a** is deposited in a zone of the removal, by which the electrode pad **113** is formed. Note that the electrode pads **114** are also formed at this time, but the electrode pads **114** are formed out of the range of FIG. **5(D)** and are not illustrated in FIG. **5(D)**.

(43) Through the processing procedure described above, the MEMS element **10** is formed.

(44) Thereafter, electrets are formed on at least one of either the fixed comb teeth **110** or the movable comb teeth **120** by a well-known electret forming method (e.g., see Japanese Patent Publication No. 5627130, etc.).

(45) The vibration-driven energy harvesting device **1** is a very minute structure processed by a MEMS technology, and lengthwise and breadthwise dimensions of the package **2** illustrated in FIG. **1** are each several centimeters, and a height dimension of the package **2** is about several millimeters.

(46) FIG. **6** is an enlarged view of a region VI in FIG. **1**, FIG. **7(A)** is a cross-sectional view taken along the line VIIA-VIIA in FIG. **6**, and FIG. **7(B)** is a cross-sectional view of a structure of a region that corresponds to FIG. **7(A)** in a comparative example.

(47) On the lead portion **112** of the fixed electrode portion **11**, the protruding portion **15** protruding toward the fixed-electrode outer periphery portion **35** is formed at a predetermined spot in an extending direction of the lead portion **112**, that is, the x-axis direction.

(48) The lead portion **112** and the fixed-electrode outer periphery portion **35** are separated from each other by the slit **16**, which is formed concurrently with the step of forming the fixed electrode portion **11** and the movable electrode portion **12** by processing the device layer **303** by DRIE, as illustrated in FIG. 4(D). The step of forming the fixed electrode portion **11** and the movable electrode portion **12** and a step of forming the slit **16** between the fixed electrode portion **11** and the movable electrode portion **12** may be performed as separate steps.

(49) In the step of forming the slit **16** by DRIE, the device layer **303** is removed entirely in a thickness direction (z-axis direction), and the insulation layer **302** is removed substantially entirely in the thickness direction. At this time, lower portions of the slit **16** side of the lead portion **112** and the fixed-electrode outer periphery portion **35**, that is, the insulation layer **302** side are undercut.

(50) FIG. 7(B) illustrates the comparative example in which no protruding portion **15** is formed on a lead portion **112**. In the comparative example, the lead portion **112** has a width, in other words, a length in the y direction being small, and as a result, a portion of the lead portion **112** at which the lead portion **112** is fixed to an insulation layer **302** is almost lost by undercutting lower portions of the lead portion **112**. As a result, the lead portion **112** may be suspended over the insulation layer **302**, or a bonding portion of the lead portion **112** to the insulation layer **302** may be broken by a minor impact, as a result of which sticking causes the lead portion **112** to adsorb or contact the fixed-electrode outer periphery portion **35**. That is, the lead portion **112** may become conductive to the fixed-electrode outer periphery portion **35**, failing to keep electric insulation.

(51) In contrast, in the structure according to the present embodiment in which the protruding portion **15** is formed on the lead portion **112**, a total width of the lead portion **112** and the protruding portion **15** is large, as illustrated in FIG. 7(A). As a result, when the lower portions of the lead portion **112** and the protruding portion **15** are undercut, an area of the bonding portion to the insulation layer **302** is kept sufficiently. Consequently, a strength of bonding of the lead portion **112** including the protruding portion **15** to the insulation layer **302** is improved, the lead portion **112** is prevented from coming into contact with the fixed-electrode outer periphery portion **35**, so that electric insulation between the lead portion **112** and the fixed-electrode outer periphery portion **35** can be kept.

(52) The embodiment described above has the following effects.

(53) The MEMS element **10** includes the base **7**, the insulation layer **8** fixed to one surface of the base **7**, the fixed electrode portions (first upper layer) **11**, at least portions of which are fixed to the base **7** and each of which includes the lead portion **112** connected to the fixed-comb-teeth connecting portion **111**, and the fixed-electrode outer periphery portion (second upper layer) **35** provided surrounding the lead portions **112** and disposed being separated from the lead portion **112** by the slits **16**, and the lead portions **112** each includes, at its predetermined portion, the protruding portion **15** protruding toward the fixed-electrode outer periphery portion **35**, and the protruding portions **15** are fixed to the insulation layer **8**. In this structure, a width of each lead portion **112** at the predetermined spot at which the protruding portion **15** is provided is large. As a result, even after lead portions **112** and the protruding portions **15** are undercut when the slits **16** are formed by etching, insulation layers necessary for fixing the fixed electrode portions **11** are left. Consequently, a strength of the fixed electrode portions **11** can be kept, so that the electric insulation between the lead portion **112** and the fixed-electrode outer periphery portion **35** can be kept.

(54) For the MEMS element **10** according to the present embodiment, it is only required that the protruding portions **15** protruding toward the fixed-electrode outer periphery portion **35** are formed at predetermined spots of the lead portions **112**, and the production method is completely the same as in a case where the protruding portions **15** are not formed on the lead portions **112**. Therefore, it is possible to improve productivity compared with a method in which protective films are formed

at spots at which undercutting is to be prevented.

(55) (Modifications)

(56) The embodiment described above exemplifies the structure in which one protruding portion **15** is formed on each lead portion **112**. However, a plurality of protruding portions **15** may be formed on each lead portion **112** along a longitudinal direction of the lead portion **112** at predetermined intervals.

(57) Further, the embodiment described above exemplifies the structure in which the protruding portions **15** protrude toward the center band portion **121** of the moveable portion electrode portion **12**. However, a structure in which the protruding portions **15** protrude in an opposite direction to the center band portion **121** of the moveable portion electrode **12** is possible. Further, in a case where a plurality of protruding portions **15** are formed on each lead portion **112**, protruding portions **15** protruding toward the center band portion **121** and protruding portions **15** protruding in the opposite direction to the center band portion **121** may be provided on one lead portion **112**.

Other Embodiments

(58) The embodiment described above exemplifies the structure in which the protruding portions **15** are formed on the lead portions **112** of the fixed electrode portions **11**. However, the protruding portions **15** may be formed on the fixed-comb-teeth connecting portions **111**. An example of a formation spot in a case where the protruding portions **15** of the fixed-comb-teeth connecting portions **111** is illustrated as a formation position **15A** in FIG. **1**. In FIG. **1**, the formation position **15A** is illustrated for only one fixed electrode portion **11**, but as a matter of course, the formation position **15A** is formed on each fixed electrode portion **11**. Further, a plurality of formation positions **15A** may be provided to one fixed-comb-teeth connecting portion **111**. A reason for forming the protruding portions **15** on the fixed-comb-teeth connecting portions **111** will be described below.

(59) The fixed-comb-teeth connecting portion **111** is bonded to the insulation layer **8** at a region **11C.sub.1** linearly extending in the y direction in the hatched zone **11C** illustrated in FIG. **3(B)**. As described in Japanese Patent Laid-Open No. 2018-88780, which is cited as Patent Literature 2, an advantageous structure for a vibration-driven energy harvesting device is a structure in which a Q factor of resonance of the movable electrode portion **12** is decreased to make a frequency of the resonance gentle, in other words, a structure that allows the resonance for a wide frequency band of outside vibration. To this end, it is necessary to decrease a parasitic capacitance that develops between the base **7** and the fixed electrode portions **11**.

(60) Therefore, a width (a length in the x-axis direction) of a bonding portion at which the fixed-comb-teeth connecting portion **111** is fixed to the insulation layer **8** illustrated as the region **11C.sub.1** in the hatched zone **11C** has to be small. However, the smaller the width with which the fixed-comb-teeth connecting portion **111** is fixed to the insulation layer **8** becomes, the more the strength of bonding between the fixed-comb-teeth connecting portion **111** and the insulation layer **8** decreases by undercutting that occurs at bonding portions between the fixed-comb-teeth connecting portion **111** and the insulation layer **8** in the DRIE processing for forming the slit **16**. Therefore, not by increasing the width of the fixed-comb-teeth connecting portion **111**, that is, not by adopting a structure that makes the parasitic capacitance large, but by forming the protruding portion **15** at a predetermined spot of the fixed-comb-teeth connecting portion **111**, in other words, by forming the protruding portion **15** partially, so as to increase the width of bonding between fixed-comb-teeth connecting portion **111** and the insulation layer **8** at the protruding portion **15**, it is possible to keep a necessary strength of bonding without increasing the parasitic capacitance even after the fixed-comb-teeth connecting portion **111** is undercut.

(61) Note that, since the fixed comb teeth **110** sides of the fixed-comb-teeth connecting portions **111** are positioned above the opening **7a** formed in the base **7**, the fixed-electrode outer periphery portion **35** cannot be provided around the fixed comb teeth **110** of the fixed-comb-teeth connecting portions **111**. However, in all regions of the fixed-comb-teeth connecting portions **111** provided on

the base, the fixed-electrode outer periphery portion **35** is provided, and there are no regions where peripheries of the fixed-comb-teeth connecting portions **111** are not covered. In the present specification, as long as the fixed-electrode outer periphery portion **35** is provided in all regions of the fixed-comb-teeth connecting portions **111** provided on the base in this manner, the structure in which the fixed-electrode outer periphery portion **35** is provided in the peripheries of the fixed-comb-teeth connecting portions **111** is considered to be included.

(62) Although the embodiment described above exemplifies the MEMS element **10** as one being formed from an SOI substrate, the MEMS element **10** may be formed from a silicon substrate instead of an SOI substrate. Alternatively, glass, metal, alumina, and the like can be used instead of a silicon substrate.

(63) In the embodiment described above, the MEMS element **10** is exemplified as a MEMS element for a vibration-driven energy harvesting device. However, the MEMS element **10** may be used for a vibration actuator that vibrates the movable electrode portion by receiving a drive voltage from the outside.

(64) Alternatively, the MEMS element **10** can be applied to a microresonator having a structure in which a movable electrode and a fixed electrode are separated from each other by a slit, as described in Patent Literature 1 (Japanese Patent Laid-Open No. 2005-323039). The microresonator described in Patent Literature 1 has a function as a filter that extracts, from among vibrations occurring between one fixed comb teeth electrode and one movable comb teeth electrode, only a specific frequency from the other fixed comb teeth electrode.

(65) Moreover, the structure of the MEMS element **10** according to the present embodiment can be applied to various types of sensors.

(66) Various embodiments and modifications have been described above, but the present invention is not limited to these details. The various embodiments and modifications described above may be combined or may be altered as appropriate, and other aspects contemplated within the scope of the technical concept of the present invention are also included within the scope of the present invention.

(67) The disclosure of the following priority application is incorporated herein by reference. Japanese Patent Application No. 2019-187152 (filed on Oct. 10, 2019)

REFERENCE SIGNS LIST

(68) **1** vibration-driven energy harvesting device **7** base **8** insulation layer **10** MEMS element **11** fixed electrode portion (first upper layer) **12** movable electrode portion (moveable portion) **13** elastically-supporting portion **15** protruding portion **16** slit **35** fixed-electrode outer periphery portion (second upper layer) **111** fixed-comb-teeth connecting portion **112** lead portion

Claims

1. A MEMS element comprising: a base; an insulation layer fixed to one surface of the base; a first upper layer, wherein at least portions of the first upper layer are fixed to the insulation layer, wherein the first upper layer comprises fixed electrode portions, each fixed electrode portion having an electrode pad, a lead portion, and a fixed-comb-teeth connection portion; and a second upper layer that surrounds the first upper layer, wherein the second upper layer is separated from the first upper layer by slits, wherein: the first upper layer includes, at predetermined portions, protruding portions protruding toward the second upper layer, wherein the protruding portions are provided on the lead portions of the fixed electrode portions at respective positions between the electrode pads and the fixed-comb-teeth connection portions or on the fixed-comb-teeth connection portions, portions of the insulation layer beneath the first upper layer are removed, the protruding portions have widths configured to fix the first upper layer to the insulation layer due to the removal of the portions of the insulation layer beneath the first upper layer, and the protruding portions are fixed to the insulation layer.

2. The MEMS element according to claim 1, wherein the insulation layer is removed in regions corresponding to the slits, and the base is exposed in a thickness direction of the insulation layer through at least some of the regions corresponding to the slits.
 3. The MEMS element according to claim 1, wherein the second upper layer is fixed to the insulation layer.
 4. The MEMS element according to claim 1, wherein the base, the first upper layer, and the second upper layer are made of silicon.
 5. The MEMS element according to claim 1, wherein the insulation layer is made of an inorganic insulation material.
 6. The MEMS element according to claim 1, wherein the first upper layer includes a plurality of fixed comb teeth, and the insulation layer is removed in regions corresponding to between the fixed comb teeth formed adjacently.
 7. A vibration-driven energy harvesting device comprising: the MEMS element according to claim 6, a moveable portion including a plurality of movable comb teeth meshing with the fixed comb teeth of the first upper layer; elastically-supporting portions elastically supporting the moveable portion; and an output portion outputting electric power that is generated by movement of electric charge occurring by relative movement between the fixed comb teeth and the movable comb teeth.
 8. A MEMS element comprising: a base; an insulation layer fixed to one surface of the base; a first upper layer, wherein at least portions of the first upper layer are fixed to the insulation layer, wherein the first upper layer comprises a fixed electrode portion having an electrode pad, a lead portion, and a fixed-comb-teeth connection portion; and a second upper layer that surrounds the first upper layer, wherein the second upper layer is separated from the first upper layer by slits, wherein: the first upper layer includes, at a predetermined portion a protruding portion protruding toward the second upper layer, wherein the protruding portion is provided on the lead portion of the fixed electrode portion at a position between the electrode pad and the fixed-comb-teeth connection portion, the protruding portion has a first width sufficient to fix the first upper layer to the insulation layer, a lead portion of the first upper layer has a second width that is smaller than the first width and larger than a third width of the insulation layer beneath the lead portion, the slits include a portion formed to follow a contour of the protruding portion, and the protruding portions are fixed to the insulation layer.
 9. The MEMS element according to claim 8, wherein the first upper layer includes protruding portions at each of multiple predetermined portions, the first upper layer includes electrode pads, lead portions, and fixed-comb-teeth connection portions, and at least one of the protruding portions is provided on one or more of the lead portions, and each of the protruding portions protrudes toward the second upper layer.
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